



Annual Report 1992-93

**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
RAJENDRANAGAR, HYDERABAD - 500 030, INDIA**



Hon'ble Union Minister for Agriculture Dr. Balram Jakhar inaugurating the ICAR special general meeting at NAARM

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ICAR

Annual Report 1992-93

National Academy of
Agricultural Research Management
Rajendranagar, Hyderabad - 500 030.

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Published by the Director, NAARM, Hyderabad

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1. INTRODUCTION



he National Academy of Agricultural Research Management (NAARM) is a national institute and a constituent of Indian Council of Agricultural Research (ICAR), which functions under the aegis of the Department of Agricultural Research and Education (DARE), Ministry of Agriculture, Government of India. NAARM is the apex body of the country working in the areas of management of agricultural research, education, extension, and training. The Academy is organized into six subject matter units, viz. Agricultural Research Systems and Policies, Educational Systems and Technology, Transfer of Technology Systems and Policies, Project Management, Human Resources Development, and Information and Documentation.

Organization of NAARM



These units are supported by centralised services which are, Administration and Accounts, Farm services including Agro Technology Demonstration Blocks, Production (Press) and Publications, Hostel Services, Hindi and Training Cells.

The major objectives of the Academy are to conduct research, and to provide training and consultancy in the areas of management of agricultural research, education, extension, and training.

THE MANDATE

- ◁ To organize and conduct training programmes in agricultural research management to scientists at various levels
- ◁ To undertake systematic review and study of management problems of ICAR research institutes, agricultural universities, agricultural research projects and systems
- ◁ To organize, liaise, and coordinate programmes of international cooperation in the field of agricultural research management
- ◁ To plan, organize, and conduct training programmes, workshops, and seminars in educational technology and structured learning for the faculty of agricultural universities and ICAR institutes
- ◁ To assist the State Agricultural Universities (SAUs) in developing regional centres of management to cater to the needs of various SAUs

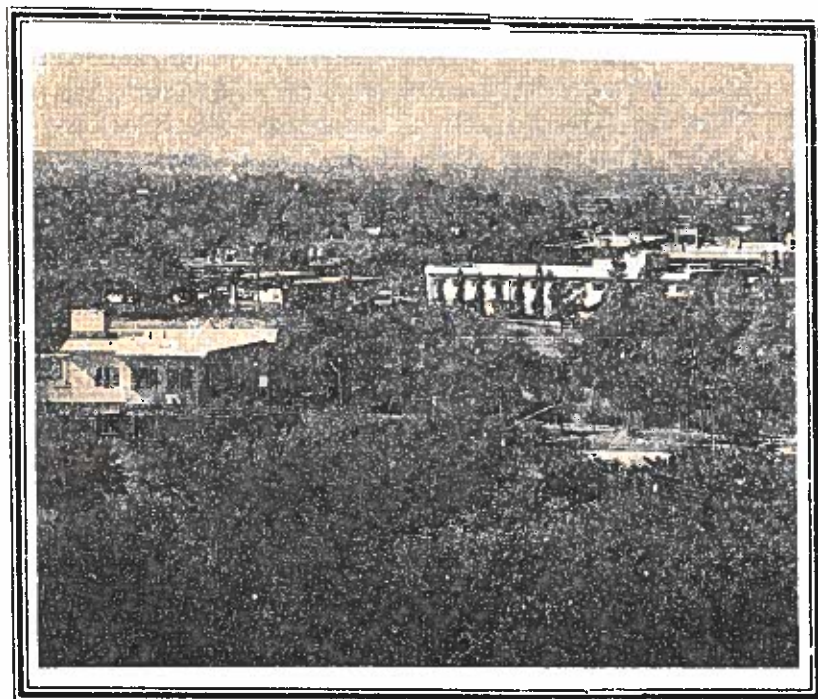
The Academy is located in the picturesque and sylvan environs of Rajendranagar in Ranga Reddy District, on the outskirts of the city of Hyderabad.

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The Academy has a beautiful campus of 50 hectares located within the Andhra Pradesh Agricultural University. The well planned campus of the Academy has several modern amenities which include the Academic Block, Faculty-cum-laboratory and Library Building, Administrative Block, Halls of Residence, Scientists Home, and several faculty and staff residences.

An Advanced Centre on Agricultural Education and Research Management, established with the support of UNDP/UNESCO/FAO, is also functioning at the Academy.



NAARM - A view

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2. ACTIVITIES



The various activities of the Academy comprise of training, research, education, extension, workshops and consultancy, all in the area of agricultural management, and each of these activities is described below.

2.1 Training

The Academy imparts training in the areas of research management, educational technology, and training management for various levels of scientists of the ICAR and the SAUs, which form a part of the National Agricultural Research System (NARS). It also conducts training programmes for executives, senior research managers, and administrative and accounts officers.

2.1.1 Foundation Courses

The most important and the largest programme of the Academy is the Foundation Course on Agricultural Research Project Management (FCARPM). This provides the foundation training to the newly recruited scientists of the Agricultural Research Service (ARS) of the ICAR. The FCARPM is a composite course which provides training in all the branches of agricultural research management. These are agricultural research systems and policies, research project management, human resources development and management, transfer of technology systems and policies, and information management.

In addition to the FCARPM, a foundation course is also offered for the finance and accounts officers directly recruited by the ICAR. This course provides detailed training in various administrative procedures such as office management and ad-

ministration, establishment, estate management, office automation, stores procurement and inventory management. This course also includes training in budget preparation, audit, finance and accounts.

During the year under report, four foundation courses in agricultural research project management and one in finance and accounts procedures were offered, which are listed in the box.

Foundation Courses				
Sl. No	Title of Programme	Date	Coordinators	No. of Participants
FCARPM				
1.	XL	Jan 21 to Jun 20, 1992	K. M. Reddy D. Rama Rao	91
2.	XLI	Jun 9 to Nov 8, 1992	P. Kumar A. Gopalani	22
3.	XLII	Aug 4 1992 to Jan 2, 1993	S. N. Saha	11
4.	XLIII	Nov 11, 1992 to April 4, 1993	V. E. Sabarathnam K. M. Reddy	53
5.	I Foundation Course for Finance & Accounts Officers	Jan 4 to Jan 28, 1993	M. Suresh Kumar R. B. Gaddagimath	9

a) Foundation Course on Agricultural Research Project Management

The course is structured into three phases, each covering a duration of seven weeks, and extending for a total period of

six months. The first and third phases are held in the campus of the Academy, whereas in the second phase, called Field Experience Training (FET), the trainees undergo practical experience at various field stations identified all over the country. The objectives of the FCARPM are listed in the box below.

OBJECTIVES

- ③ To acquaint the scientists with the general agricultural scenario in the country and the world with reference to research and development
- ③ To equip them with techniques in research project management
- ③ To develop skills in organizational behaviour for high motivation, team work, and conflict resolution
- ③ To give insight into the rural society and structure
- ③ To provide training for communication in agriculture and transfer of technology
- ③ To orient the scientists to the ARS of ICAR and the national agricultural research, education and extension systems in the country
- ③ To foster a feeling of fraternity among the scientists serving in different parts of the country

During the first phase of the training, the scientists are exposed to various aspects of general agricultural scenario, national and international agricultural research systems, project management, organizational behaviour, transfer of technology, information management, and administrative and accounts procedures. The scientists are also provided with adequate opportunities to have hands-on experience for developing skills in computers, audio-visual aids, organizational behaviour, video, accessing information, treatment of research findings, and communication skills. With the exposure to the various aspects of problem identification and project formulation during the first phase, the trainees proceed to the Field Experience Training (Second Phase) to various Regional Research Stations of State Agricultural Universities and selected ICAR Institutes.



A FCARPM Valedictory Function

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During FET, they get a first hand experience on problem identification, technology development process, and mode of transfer of technology. They also carry out a socio-economic survey of villages to get a feel of the rural life and the existing indigenous knowledge systems. This phase provides the trainees an opportunity to prepare themselves to work with the farmers, thereby enabling them to develop need based projects. The third phase of the programme gives an opportunity for the scientist trainees to share their field experience and also the rationale for developing a viable and meaningful research project using various project management and related techniques learnt during the first phase. The third phase also enables the trainees to develop an appreciation for interpersonal relationships and for working in groups, thereby enabling them to work in interdisciplinary project teams for conducting research to solve the problems of farmers.

The overall programme also includes the visit of the scientist trainees to various research institutions in and around Hyderabad to familiarize them with different organizations and their research activities. The programme has an in-built as well as terminal evaluation to assess the extent to which the trainees have imbibed the knowledge and skills imparted during the course of the training programme.

b) Foundation Course for Finance and Accounts Officers of ICAR

This foundation training course for the newly recruited finance and accounts officers of ICAR is aimed at providing an overview of the general financial accounting system in ICAR, and other management principles. This programme helps to improve the skills and efficiency of finance and accounts officers in discharging their functions effectively, which in turn would help them in extending efficient financial and accounting support to the Directors of the institutes to achieve the organizational goals and needs. Various areas such as financial and accounting system, work simplification, retirement benefits and pension schemes, budget systems,

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delegation of financial powers, leave rules and other personal benefits, and labour laws were covered during the course, in addition to providing a general orientation to the trainees on various aspects of human resources management, library management, computer applications in accounting, and central administrative tribunal.

2.1.2 Senior Level Courses

Since research management training is a continuous process of learning and updating different skills, the Academy offers various senior level programmes to different categories of scientists. These programmes essentially aim at developing an appreciation and providing adequate skills to various levels of scientists such as Principal Scientists, Professors, Heads of Divisions/Departments, Directors of Research, Project Coordinators, Directors and Deputy Director Generals of ICAR and SAUs. During the year under report 12 senior level programmes were offered which are enlisted in the box. A brief account of these various courses is presented below.

a) Management Information System for Agricultural Research

This course provides guidelines on the methodology for the development of computer based Management Information System (MIS), and Project Budgeting System (PBS) that would ultimately help managers to take timely and meaningful decision on various aspects of research projects, human resources, and financial resources. The course enables the participants to develop skills through project work, case study, and hands on experience with personal computers.

b) Agricultural Research Project Management

This course aims at exposing research managers to various concepts of agricultural research projects, and application of management techniques for efficient implementation of research projects in order to make them relevant.

Senior Level Courses

Sl. No.	Title of the Programme	Date	Coordinators	No. of participants
1.	Management Information System for Agricultural Research	April 13-24, 1992	T. Balaguru J. C. Kalla	8
2.	Agricultural Research Project Management	April 21-May 1, '92	S. N. Saha	35
3.	Participatory Video for Social Change	June 16-27, 1992	R. Chithiraichelvan S. Shanmugam	11
4.	Programmed Instruction with Special Reference to Video and Computers	June 30-July 10, 1992	A. Gopalam K.M. Reddy	11
5.	Computer Applications in Agricultural Research	July 14-24, 1992	U. Muralidhar	27
6.	Planning and Management of Video based Instructional System	Aug 3-14, 1992	A. Gopalam K. R. Prabhakar	20
7.	Training of Trainers	Aug 17-27, 1992	T. Balaguru P. Manikandan	17
8.	Management of People at Work	Aug 18-29, 1992	J. Challa M. M. Anwer	8
9.	Administrative and Financial Management	Sep 1-5, 1992	V. Raghavendra Rao	6
10.	Management Information System for Agricultural Research	Sep 7-19, 1992	P. Manikandan T. Balaguru J. C. Kalla	9
11.	Computer Application for Information Storage and Retrieval in Agricultural Libraries and Information Centres	Sep 28-Oct 9, 1992	R. B. Gaddagimath	27
12.	Executive Development Programme on the Management of Human Resources with a Focus on Women Managers	Nov 17-28, 1992	M. M. Anwer S. Shanmugam	9

Various aspects of research project cycle and other related topics like management of differences and brainstorming were covered through information sharing, group discussion, case analysis, and exercises.

c) Participatory Video for Social Change

The course exposes the scientists to the participatory approaches to communication and message development, and enables them to use video as a medium of documentation and development. An insight is also provided on the dynamics of human communication in the production of video. Various areas related to participatory approaches and the use of video as a tool for social change were covered in addition to human behavioural aspects, through a mix of theoretical and hands on activities.

d) Programmed Instruction with Special Reference to Video and Computers

The main objective of this programme is to provide an orientation to programmed learning process and also to develop an understanding on instructional carriers and different teaching gadgets. Theoretical orientation to the basic aspects were reinforced through hands on experience for using video and computers as instructional aids in learning and teaching process, and in programmed learning setting.

e) Computer Applications in Agricultural Research

This course aims at providing skills to the senior scientists of ICAR institutes on the use of computers and different software packages for processing of data and presentation of meaningful information. Various user - friendly packages related to word processing, database analysis, electronic spread sheet, and statistical analysis were covered in addition to providing a brief account on computer virus and vaccines. Major emphasis in the pedagogy was laid on hands on experience with computers for the use of various packages.

Coordinators	No. of participants
T. Balaguru J. C. Kalla	8
S. N. Saha	35
R. Chithiraichelvan S. Shanmugam	11
A. Gopalam K.M. Reddy	11
U. Muralidhar	27
A. Gopalam K. R. Prabhakar	20
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J. Challa M. M. Anwer	8
V. Raghavendra Rao	6
P. Manikandan T. Balaguru J. C. Kalla	9
R. B. Gaddagimath	27
M. M. Anwer S. Shanmugam	9

f) Planning and Management of Video Based Instructional System

The course aids the participants to develop video programmes by utilizing the modern technologies, thereby helping to clear the myth of complexities associated with video programme production. Three programmes, viz. use of 'Neem coated ureas increase paddy yields', 'Composite fish culture', and 'Cattle and buffalo breeds' were prepared during the course of the programme.

g) Training of Trainers

This programme helps the teachers and trainers to improve their knowledge and develop certain specialized skills in order to organize more effective training programmes. The programme consisted of a blend of lectures, exercises, group discussion, and practice sessions to cover various aspects such as basics of training, role of trainers, and training design, delivery and evaluation.

h) Management of People at Work

Keeping in view the importance and usefulness of effective management of human resources, this course is organized to develop an understanding of human behaviour among the participants with a view to change, direct, and control their behaviour. Emphasis is laid on group behaviour in addition to individual behaviour. The programme used a rich variety of training methods such as group discussion, case analysis, experiential learning, practicals with video feedback, and the lecture method.

i) Administrative and Financial Management

This programme is designed for the Directors of the institutes and Assistant Director Generals working at ICAR headquarters, with a view to generate and discuss various issues related to administrative and financial management.

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Issues related to administrative efficiency and productivity, financial and labour management, vigilance and disciplinary issues, and budget and resources allocation were considered during the deliberations.

b) Computer application for Information Storage and Retrieval in Agricultural Libraries and Information Centres

The course facilitates the participants to recognize the effectiveness of computers for the library information management and also help them identify the areas for computerization to tackle the problems of abundant information. Main emphasis was given on the use of UNESCO's CDS/ISIS (version 2.32) for information storage and retrieval through theoretical orientation, hands on experience with computers and exercises.

k) Programme on the Management of Human Resources with a focus on Women Managers

Considering the increased role and importance of women



Inaugural Function - Chief Guest, Dr. Girja Vyas, Hon'ble Dy. Minister for Information and Broadcasting. Dr. J.C. Kalla, Director addressing the participants

managers in agricultural sector, this course is organized to expose the women managers to organizational behaviour, human resources management, and organizational development. In addition to the coverage on general aspects of the management of human resources, special areas of interest to women managers such as changing role of women managers, women as successful managers, issues in organizational behaviour with exercises, and group discussion were also covered.

2.2 Research

The Academy is the premier institution of the country researching in the area of agricultural management. The academy carries out research in various aspects such as agricultural policy, research project management, transfer of technology, human resources management, agricultural education, and information technology. During the year under report, 10 research projects were under investigation. A brief description of the research projects is presented below :

2.2.1 Development of database on crops, forestry, animal and fishery resources in the national agricultural research system (K.Vidyasagar Rao and P.Kumar)

The project aims at documenting and developing databases on crops, fisheries and livestock. These databases, once developed, would be very useful to both planners and administrators. This would also be useful in updating the teaching and training materials used in the various programmes of the Academy.

The relevant parameters of animal and fishery resources have been identified and data are being collected and compiled. The data collected include trends in the production of milk, wool, egg and livestock at the country level as well as at the global level. Data are also collected on export oriented marine products.

2.2.2 Assessment of returns to expenditure on research and development of production technologies for rice fallow pulses and chillies in Andhra Pradesh (K.Purnachandra Rao)

Data were collected on the performance of the rice fallow blackgram and chillies in 1992-93 from the Lam Research Station in Andhra Pradesh. The collected data were analyzed and the report writing is in progress.

2.2.3 AGRIUNIS (D.Rama Rao and U.Muralidhar)

The project aims at collecting data on manpower, project, research, and academic activities of all the agricultural universities in the country. Dr.Rajeshwari of the Department of Science and Technology is the Joint Adviser of the Project.

The advisory committee of the project met on May 15, 1992 and decided on the schedules for data collection and also on the methodology to be used in the project. The expectations on the outcome of the project were also discussed. Following this, a software package was developed for handling the data that are expected to be collected. All the agricultural universities were requested to nominate coordinators for the AGRIUNIS Project and the first coordinators' meeting was held on July 28, 1992 to apprise them the methods of data collection; the financial and technical support needed for this purpose by the universities were also discussed. The schedules for data collection were discussed in detail and the final version was approved. After printing the schedules, efforts are on for data collection from different universities. The data collected will be aggregated on activity basis, viz. Academic, Research, Extension, and Administration.

2.2.4 Development of a suitable Management Information System (MIS) for the NARS (T.Balaguru, P. Manikandan, Jagdeesh C.Kalla and K.Vidyasagar Rao)

The project was undertaken with the following objectives:

- * to develop a suitable MIS to serve the research managers in the research system as a useful tool for managing their research activities, human resources, funds and other physical

facilities more efficiently and effectively;

- * to familiarize the decision makers in the NARS with the potentials and use of a well-organized MIS; and

- * to generate efficient inferential system from the database.

Two more case studies were prepared under this continuing project. Through mailed questionnaires and personal visits, the required data on projects, personnel, funds, and facilities were collected from two of the regional agricultural research stations of the Tamil Nadu Agricultural University located at Vriddhachalam and Paiyur. The collected data were consolidated and the MIS for these two stations were developed in the form of case studies. The developed cases were sent to the respective station heads for their comments and feedback. Efforts are on to publish the basic MIS guidelines including the case studies, utilizing the funds provided by ISNAR, for wider circulation among the ICAR institutes and SAUs.

2.2.5 Evolving a suitable methodological framework for priority setting in agricultural research (Jagdeesh C.Kalla and T.Balaguru)

In order to assist the policy makers and the research managers to assess the actual research needs and to set priorities appropriately, the project was undertaken with the following objectives :

- * to study the state of art of research prioritization in the NARS;

- * to analyze the suitability of various research priority setting methods in the research system at strategic and at programme/project levels;

- * to reconcile and evaluate the results of alternative priority setting methods for research at these levels; and

* to elicit policy implications from the most optimum view vis the existing pattern of research prioritization in agriculture.

The basic questionnaire developed for data collection was modified on the basis of feedback received from the scientists at the National Research Centre for Camel and Assam Agricultural University. The revised questionnaire would be used for further data collection. Some basic information like contribution of different commodities to agricultural GDP, and various critical factors used in research prioritization have been collected.

2.2.6 Communicative openness among scientists in agricultural research organizations (R.K.Samanta and Jagdeesh C.Kalla)

The study was initiated with the following objectives :

- * to find out the perception of scientists of ICAR about their communicative openness;
- * to find out the perception of ICAR scientists about the work environment relating to job satisfaction; and
- * to study the degree of satisfaction of the scientists of ICAR with the sub-components of organizational communication in their organizations. The relevant data were collected through structured questionnaires. The analysis and the interpretation of the data are in progress.

2.2.7 Culture of Trust (M.M.Anwer, Jagannadham Challa and S.Shanmugam)

Work on culture of trust in research organizations was continued during the period to determine the extent of interpersonal trust/distrust among the research scientists in different ICAR institutes, to identify the causes of distrust among the scientists, and to arrive at alternative actions to develop trust among scientists. Collection of required data for the project was completed; analysis and report writing are in progress.

2.2.8 A profile of scientific and technical human resources in NARS (S. Shanmugam and Jagdeesh C.Kalla)

This study aims at analyzing the socio-economic contributions to cadre strength of the academic, scientific, and technical personnel of ICAR and SAUs. It also deals with the scientific and technical staff interface in relation to projects undertaken by them. This project will help in planning optimum human resources profile in the NARS for 2001 AD. Collection of data on scientists of ICAR was initiated during the year under report.

2.2.9 Management of agricultural research stations (K.Venkateswarlu)

The study aims at understanding the development process of agricultural research stations; management of land, labour, capital, crops, irrigation systems, fertility management, greenhouses, gardens, workshops, and research projects; transfer of technology; and operational constraints. Several case studies were conducted under this project. The cases developed were used in the workshop conducted for managing agricultural research stations.

2.2.10 Forecasting of sorghum scenario in India by 2000 AD (D.Rama Rao and V.R.Kiresur)

This research project was conducted to forecast the sorghum scenario in India by 2000 AD using the techniques of futurology. Dr. V.R.Kiresur, Scientist from Directorate of Oilseeds Research, Hyderabad was an associate in the project. Using the sorghum production and crop area data from 1950 to 1990, the forecasts were made by trend extrapolation as well as by Delphi methods. The following are the salient findings of this project :

- i) The data on area and production of sorghum show a scatter and the forecast too show a scatter of about 10%.
- ii) The variation on the per cent crop area under high yielding varieties is relatively smooth and shows an increasing rate of change in the recent past.

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smooth and shows an
recent past.

iii) The forecasts by trend extrapolation and Delphi are comparable in case of total crop area, and yield.

Forecast of Sorghum Scenario by Trend Extrapolation and Delphi

Parameter/Event	Actual 1990-91	Forecast for 2000 AD Trend	Delphi
Area (million ha)	14.50	12.30	13.00
Production (million tonnes)	11.88	13.10	12.28
Yield (Kg/ha)	819.00	1065.00	1000.00
Per cent area under HYVs	52.10	67.50	62.50
Relative procurement price with reference to rice and wheat			
Sorghum/wheat	0.84	0.86	--
Sorghum/rice	0.88	0.88	--
No. of years for sorghum to become attractive for alternate uses	--	--	10

iv) The forecast for the area under high yielding varieties is different as estimated by the two methods; the trend extrapolation is higher by about 7.5% as compared to Delphi forecast.

v) In order to increase sorghum production, the support price has to be relatively higher than the present one.

vi) Any breakthrough in dryland crops will have severe impact on sorghum scenario, as such crops are likely to grow at the expense of sorghum.

2.3. Education

Educational programmes in the State Agricultural Universities have to reflect the arising needs and demands in

every aspect of education, viz. curriculum development, instructional design and methods, evaluation and assessment, and educational administration. There is a need to develop suitable technical manpower in agricultural universities, who are serving and are specialists in their own subject, trained in all these aspects of agricultural education. With a view to orient and train them in all the areas of pedagogy relevant to higher education in agriculture, a programme in educational technology, using a distant mode of education coupled with three contact programmes, is being conducted by the Academy. Four Senior Associate Professors from the various SAUs graduated from the first programme which ended by March, 1993. The programme was coordinated by Dr.A.Gopalam.

The programme is of one year duration with 46 modules; each module deals with specific class-based issues under relevant curricular area of study. The module engages the learners through interactive processes of analysis and synthesis in terms of practice-based principles, and attempts to impart improved teaching competencies among the learners. These modules are organized in five learning packages. The response sheets of each module, when completed and evaluated, facilitate the learners to get their learning reinforced.

The programme has three contact sessions each of two weeks duration. The first contact session helps to orient the learners to distant learning process and related experiential activities. Emphasis is given on group discussion and practical exercises during the second and third contact programmes. A description of the various aspects covered in the three contact sessions is given below :

Contact Session 1

1. Goals and objectives of the OPET
2. Discussion and refinement on design of curriculum for the OPET
3. Discussion of instructional module and design of assignments
4. Project work

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5. Methods of time management and additional re-
source materials for carrying out assignments
6. Production of resource materials for classroom use
and experimentation; hands-on experience in the
use of slide projector and overhead projector (OHP)
materials
7. The functional orientation and organization of con-
tact sessions
8. Introduction to the tutorial/ward system
9. Core books and journals for reference and for fur-
ther autonomous learning
10. Evaluation system and its operation

Contact Session 2

a. Group discussion (for group learning)

1. Macro-curriculum of an undergraduate programme,
credit distribution, sequential organization, etc.
2. Use of episodes, metaphors and analogies
3. Study tours as means of development education
4. Village stay programme as experiential learning
process
5. Inculcation of corporate culture and values among
students
6. Interdisciplinary courses and team teaching
7. System of teaching : recurrent and residual issues

b. Practical exercises

1. Modifying a course into performance objectives
2. Drawing an instructional plan for a classroom inter-
action
3. Structuring a lecture and micro-teaching
4. Preparing slides, OHP materials, and charts for a
class
5. Preparing worksheet for field-based problem solv-
ing
6. Preparing laboratory manuals and audio-tutorial
lessons
7. Preparation of competency-based question paper
for mid-term/final examination for a course

Contact Session 3

a. Group discussion (for group learning)

1. Learning styles of the slow learners and their improvement
2. Global trends and perspectives in undergraduate agricultural education
3. The interaction of science, technology, and society
4. Issues and strategies for faculty improvement
5. How to make postgraduate education more effective?
6. Creativity
7. Non-monetary inputs for improving teaching-learning

b. Practical exercises

1. Building a curriculum for a course (topic-wise, performance-wise, and activities-wise for an interdisciplinary course)
2. Preparation of an experiential information processing model for learning-teaching of a unit/lesson for a video programme
3. Preparing for a poster session and demonstrating its use for integrating research, education, and extension education
4. Preparing a programmed lesson or review for self-instructional purposes
5. Script writing and shooting for an educational video lesson
6. Computer-assisted lesson production
7. Brainstorming and dendrogramming

In addition to the various modules and contact sessions, each participant undertakes two projects, one class room-based and another institution based, during the course of the programme. Each trainee is also required to take up individual study on an educational topic related to higher education in order to offer a seminar during the third contact session. Successful candidates are awarded a postgraduate diploma in educational technology.

2.4 Extension

The Agro-Technology Demonstration Blocks (ATDB) constitute the extension activities of the Academy. ATDB, established in an area of 30 hectares, is conceived as the show window of the NARS. In this demonstration, an attempt is made to put on show the latest agricultural technologies developed by the SAUs and ICAR institutes. The demonstrated technologies are those which are suitable to the agro-climatic conditions of Hyderabad.

The ATDB has three major sections which are demonstration of horticultural crops, field crops, and sylvi systems. The demonstration of horticultural crops consists of fruit orchards, vegetables, and commercial floricultural crops. The fruit orchards comprise of 0.5 ha of mango, and ber; 0.25 ha each of grapes, guava, sapota, and minor fruit crops like fig, pomegranate, etc. The varieties under demonstration in the mango plot are Amrapali and Mallika, the hybrids released by the Indian Agricultural Research Institute; and Ratna, the hybrid released by the Konkan Krishi Vidyapeeth; hybrid 10 and 13 of the Indian Institute of Horticultural Research. Some of the established popular commercial varieties like Benishan, Dasher, Peddarasam, etc. are also grown. The guava varieties under demonstration are Allahabad Safeda, Lucknow 49, Banarsi Surkha, Safed Jaam, and Kohir Safeda.

The sapota plot is planted with the varieties such as Cricket Ball, Kalipatti, Oval, and Bombay. The varieties being demonstrated in the ber plot are Umran, Gola, Seb, and Mundaya. In the vineyard the grape varieties being grown include Anab-e-Shahi, Thompson Seedless, Dilkush, Kalisabibi, Arka Kanchan, Arkavati, Arka Shyam, Arka Hans, etc.

The pomegranate varieties demonstrated in the minor fruits plot are Jyothi, released from the University of Agricultural Sciences, Bangalore, and Ganesh from Mahatma Phule Krishi Vishwavidyalaya, Rahuri.

In the vegetable demonstration plot five kinds of vegetables were grown both during *kharif* (27 varieties) and *rabi* (13 varieties). The major activity in the floricultural demonstration plot is the production of cut flowers, 0.25 ha devoted to jasmine and 0.12 ha to rose production.

About 8.5 ha of land is earmarked for the growing of field crops. In summer 1992, a demonstration was laid out with the multi-cut fodder varieties of sorghum. During the *kharif* eight varieties of four different crops were cultivated.

In addition to the regular crop demonstration, the ATDB has non-conventional energy harnessing systems such as domestic bio-gas unit, solar water heater system, and wind mill. The Agro-technology demonstration blocks concept is an unique innovation where the technologies released by different institutions are demonstrated at one place for the benefit of national and international visitors. All the scientist visitors and farmers passing through the Academy have an opportunity to view these demonstrations throughout the year. During the year under report about 1000 visitors had an opportunity to view these demonstrations.

The Academy won several prizes at the Rose Shows conducted by the Horticultural Society, Secunderabad and Hyderabad Rose Society. In the Rose Show conducted by the Horticultural Society on December 12, 1992 the Academy won the Queen of the Show Trophy, Jagdish Prasad Sonthalin Challenge Cup and several other first and second prizes under various categories. The Academy won the coveted Indira Memorial Silver Trophy for the best collection of white roses in the show held by the Hyderabad Rose Society on December 25, 1992.

2.5 Workshops

The Academy conducts a number of workshops to provide a forum for exchange of information and interaction among the scientists of the National Agricultural Research System. These workshops address the areas of current and upcoming interest in the field of agricultural management. During the year under report eight workshops were conducted, the details of which are given in the box.

Workshops				
Sr. No.	Title of the Programme	Date	Coordinators	No. of Participants
1	Curriculum Design and Development	May 4-5, 1992	Devesh Kishore	13
2	Communication Research Methodology	June 22-30, 1992	Devesh Kishore	20
3	Meaningful Learning as a Communication Process	July 7-10, 1992	R. K. Samanta	35
4	Management of Agricultural Research Stations - Operations Management	Sep 22-25, 1992	K. Venkateswarlu	26
5	Planning and Utilization of Scientific Manpower in Agricultural Research System	Oct 12-16, 1992	VE. Sabarathnam	11
6	Technology Forecasting in Agricultural Research	Nov 18-21, 1992	D. Rama Rao	14
7	Technology Generation and Transfer for Development	Nov 18-21, 1992	A. U. Gadewar R. Chithiraichelvan	29
8	Management Information System for Agricultural Research	Jan 5-6, 1992	J. C. Kalla T. Balaguru	9

A brief description of these workshops is given below :

2.5.1 Workshop on Curriculum Design and Development

The objective of the workshop was to provide an exposure to the Directors of Summer Institutes of ICAR on the principles of educational technology and curriculum development. The summer institutes are designed to bring about a qualitative improvement by updating teachers, researchers, and extension specialists with the latest knowledge and techniques in the respective fields of their specialization. The design of this workshop specifically provided a forum of interaction between co-professionals to exchange experiences, obtain feedback, and to enhance overall relevance of educational technology and curriculum development. The workshop also trained the participants in the designing of a new course or a set of courses by restructuring existing courses into the modern formats of educational technology.

2.5.2 Workshop on Communication Research Methodology

This workshop on Communication Research Methodology was organized in collaboration with the National Council of Development Communication. The main objective of the workshop was on conceptualization and standardization of the communication and its utilization. The workshop provided an excellent opportunity to the participants to have hands-on-experience in developing communication research projects, which included video and computers as communication tools. The participants consisted of faculty members from SAUs, scientists of ICAR institutes, and media personnel connected with communication research.

2.5.3 Workshop on Meaningful Learning as a Communication Process

The workshop was designed to stimulate thinking on teaching and learning from a communication perspective. It addressed the issue of creating appropriate environments and organizing content resources for greater interaction in the classroom, ultimately making it possible to achieve excel-

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lence in teaching and learning. The workshop itself was a model of process / experiential learning where the teacher acts as a facilitator as against didactic teaching methods. It demonstrated that effective communication processes create meaningful learning experiences thereby serving as a tool for bringing about change in human understanding and behaviour. The participants were faculty members from various ICAR institutes and SAUs.



Dr.J.C. Kalla, Director distributing the certificates to the participants

2.5.4 Workshop on Management of Agricultural Research Stations: Operations Management

The main objectives of the workshop were to identify the important factors for establishing a research station, and to

plan its development. This was achieved by discussing the cases on the following Agricultural Research Stations, viz. College Farm, Andhra Pradesh Agricultural University (APAU), Rajendranagar; Regional Agricultural Research Station (RARS), APAU, Anakapalli; RARS, APAU, Lam; RARS, Kerala Agricultural University, Kumarakom; Central Rice Research Institute, Cuttack; Plant Quarantine Regional Station, NBPGR, Hyderabad; and Central Institute of Fresh-water Aquaculture, Kausalyaganga. Associate Directors of Research of SAUs and Principal Scientists of ICAR institutes involved in farm management, attended the workshop and discussed the various issues related to the management of agricultural research stations with special reference to operations management.

2.5.5 Workshop on Planning and Utilisation of Scientific Manpower in Agricultural Research System

The workshop aimed at improving the understanding and developing skills among the participants in scientific manpower planning and forecasting, taking into consideration the various parameters that affect the manpower efficiency. Scientists and research managers from ICAR and SAUs participated in this workshop. During the course of the workshop an efficiency coefficient was developed in respect of scientists. In addition, nine indicators of efficiency for scientists were also identified.

2.5.6 Workshop on Technology Forecasting in Agricultural Research

This workshop, organized in collaboration with Technology Information Forecasting and Assessment Council (TIFAC), New Delhi, provided an exposure to the agricultural research scientists to the principles of technological forecasting and its application in agricultural research. Case studies in agriculture, veterinary, and agricultural engineering disciplines developed by various experts were utilized for this purpose. The workshop was attended by senior faculty members and scientists, who are involved in research management, planning, policy making, and resource allocation.

2.5.7 Workshop on Technology Generation and Transfer for Agricultural Development

The workshop was organized with the objective to develop a set of guidelines and tools for the practical use in assessing the total or the components of the technology generation and transfer systems. This was achieved by acquainting the participants with different approaches to the systems of agriculture technology generation and transfer, and by incorporating the experience of participants to identify the constraints for suitable modification. Group discussions were held on Implication Issues in Farming Systems Research/Extension, Participatory Approaches and Indigenous Technical Knowledge (ITK), and Research-Extension Linkages. The increasing services of Non-Government Organizations (NGOs) in transfer of technology were recognized and a need to formulate a collaborative and participative strategy was stressed.

2.5.8 Workshop on Management Information System for Agricultural Research



Dr.T. Balaguru welcoming the delegates; Dr.J.C.Kalla and Dr.P.Manikandan are also seen.

The objectives of the workshop were to sensitize and familiarize the senior level functionaries in the NARS for a computer based MIS and also to work out ways and means of implementing it. A variety of learning techniques were used for covering the three major themes which are guidelines on MIS, development of a project budgeting system, and use of keywords. Specifically identified issues were taken up for discussion in the workshop. The participants were Directors from selected ICAR institutes and Directors of Research from SAUs.

2.6 Consultancy

The Academy provides consultancy services in the area of Agricultural Research Management and Educational Technology. The Academy guides the SAUs and ICAR institutes in conducting training programmes in these areas. The faculty members of the Academy, specialized in various sub-disciplines, deliver special lectures in institutions which organize training programmes for their own staff. These institutions find the participation of NAARM faculty immensely useful.

The following consultancy services were provided by the Academy during the period under report.

2.6.1 Consultancy to Assam Agricultural University (AAU)

At the request of the Vice Chancellor and Director of Extension of AAU, and the Agricultural Secretary of the Government of Arunachal Pradesh, a condensed one week NAARM - AAU travelling workshop on agricultural research management was organized for the senior teachers, scientists, and developmental officers of the North Eastern States of the country. Dr.T.Balaguru and Dr.P.Manikandan of the Academy were the resource persons for this travelling workshop.

2.6.2 Consultancy to Project Management Unit of the Indo-Dutch Project

Dr.M.M.Anwer and Dr.Jagannadham Challa provided consultancy for the production of training materials, and a

training module entitled leadership and its challenge to the national water management project and managers was developed. This module is for training the irrigation engineers of the entire country.

2.6.3 Consultancy to NCERT

In order to help the National Council for Educational Research and Training (NCERT) to transform four agricultural based courses into modules, consultancy was provided by the HRD Unit and a workshop was organized. This was under the scheme of reorganization and vocationalization of agricultural education. Dr.M.M.Anwer was the Director of the workshop, and Dr.P.Kumar and Dr.Jagannadham Challa of the Academy were resource persons.

2.6.4 Consultancy to the Institute of Cooperative Management, Rajendranagar, Hyderabad

The Institute of Cooperative Management sought the consultancy of the HRD unit to review its organizational behaviour component of the training in cooperative management. Dr.M.M.Anwer, Dr.Jagannadham Challa and Dr.S.Shanmugam participated in this review workshop on 10th July, 1992.

2.6.5 Consultancy to Animal Welfare Board of India

Consultancy was provided to the Animal Welfare Board of India for conducting animal welfare training programmes in Andhra Pradesh. Dr.Jagannadham Challa acted as the chief resource person and Dr.P.Kumar, Dr.M.M.Anwer and Mr.M. Suresh Kumar were the other resource persons from NAARM in the two training programmes conducted at the Academy. Dr.Jagannadham Challa was the chief resource person for the programmes, one each conducted at Nizamabad, Kurnool, and Tirupathi. The Royal Society for the Prevention of Cruelty to Animals of the U.K. were the co-sponsors of the training programmes.

2.6.6 Consultancy to other Training Institutions

The Academy provided expertise in the shape of guest lectures to other local training and educational institutions such as the Extension Education Institute, and the Home Science College of the Andhra Pradesh Agricultural University, S.V.P. National Police Academy, Institute of Cooperative Management, Regional Station for Forage Production and Demonstration, International Crops Research Institute for the Semi Arid Tropics, MANAGE, State Bank Institute of Rural Development, WALAMTARI, NISIET, and Human Resource Development Division and R & D Unit of BHEL. Dr.P.Kumar, Dr.T.Balaguru, Dr.K.P.C.Rao, Dr.M.M.Anwer, Dr.Jagannadham Challa, Dr.S.Shanmugam, Dr.R.K.Samanta, Dr.P.Manikandan, and Dr.A.Gopalam, faculty members of the Academy, were invited to deliver guest lectures in the above mentioned institutions; subjects covered in these lectures were from the different areas of management and educational technology.

2.7 Miscellanea

2.7.1 ICAR Special General Meeting

The Special General Meeting of the ICAR Society was held at the Academy on August 30, 1992; for the first time outside New Delhi. Agricultural ministers from eleven states and fisheries ministers from three states attended the meeting. Five Parliament members and several high dignitaries of both central and state governments attended the meeting. The meeting was chaired by the Hon'able Union Minister of Agriculture Dr. Balram Jakhar. Mr. K.C.Lenka, Hon'able Union Minister of State for Agricultural Research and Education was also present. Dr. V.L. Chopra, Director General, ICAR welcomed the delegates. Deliberations were held on several key issues like price hike of fertilizers and its impact on agricultural production, coordination between research and extension, training of farmers etc. Dr. Jakhar dedicated the Faculty cum Laboratory Building of the Academy to the Nation on this occasion.

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2.7.2 All Academy Guest Lecture

Mr.Surya Prakash, Artist-in-residence at the Centre for Cellular and Molecular Biology as well as at the L.V.Prasad Eye Institute presented a slide show on the "Evolution of Indian Contemporary Art" on March 16, 1993 for the trainee scientists of the 43rd FCARPM, faculty members, officers, and staff of the Academy.

2.7.3 Anti-terrorism Week

The last week of May 1992 was observed by the Academy as Anti-terrorism week. On May 29, 1992 the trainee scientists organized an elocution competition on the theme of Anti-terrorism. A blood donation camp was organized in collaboration with the Indian Red Cross Society on May 30, 1992. Thirty six staff members and scientist trainees donated blood on this occasion.

2.7.4 Hindi Day

The Academy celebrated Hindi Day on September 16, 1992 under the Chairmanship of Dr Jagdeesh C.Kalla, Director, NAARM. Mr. C.R.Ramachandran, Assistant Director, Hindi Teaching Scheme and Incharge, Central Hindi Training Sub-Institute, Hyderabad was the chief guest. Dr.Jagdeesh C.Kalla. in his presidential address, stressed the importance of link language in a country like India and observed that Hindi can eminently play this role. Mr C.R.Ramachandran emphasized that though the literary form of Hindi is used by scholars, the language should be simple and easy to comprehend for popularizing its use in the regular official work. Dr.A.Gopalam, Officer-in-charge for Hindi explained the various activities of the Hindi Cell in promoting the language. Mr.D.Venkateswarlu, Technical Officer (Hindi) welcomed the gathering and Mr Pradeep Singh, Hindi Translator proposed the vote of thanks.

2.7.5 Women in Agriculture Day

Women in Agriculture Day was celebrated at the Acad

emy on December 4, 1992. A panel of four scientists from the Home Science discipline were invited to interact with the women workers of the Academy. Dr (Mrs) O.R.Radha Krishna, Mrs.N.Vani Reddy, Dr.(Mrs) Sarojini and Dr. V.Vimala, who are experts in the areas of human development and family studies, and foods and nutrition were invited for this purpose. These specialists were drawn from the All India Coordinated Research Project (AICRP) on community development, AICRP on Food and Nutrition, and the College of Home Science.

About 15 women working in different units of the Academy had an opportunity to interact with this panel on different issues concerning the nutrition, human development, and family welfare. The discussion focussed mainly around educating the women on proper feeding of children, and expectant mothers, and also on the avoidance of discrimination between male and female child.

2.7.6 Sadhbhavana Diwas

The Academy observed Sadhbhavana Diwas on August 20, 1992. The staff members assembled and took a pledge to preserve the harmony and unity of the nation.

2.7.7 NAARM - The Runners-up in Table Tennis

The NAARM contingent of sportsmen participated in the ICAR Inter-institutional Sports Meet held at the Central Marine Fisheries Research Institute, Cochin from November 5-9, 1992 and secured second place in table tennis in both singles and doubles events. The singles match for the second place was won by Mr.Sham Bahadur and the doubles team comprised Mr.Sham Bahadur, Mr.S.Swamy and Dr.Gaddak.



3. FACILITIES AND SERVICES



The Academy has established several facilities for the conduct of training, research, education, and extension activities in the area of management of agriculture. The academic activities are further supported by centralized services.

3.1. Facilities

3.1.1 Academic Block

The Academic Block is a modern building consisting of four well designed class rooms, each with the capacity to accommodate 60 trainees. These rooms are equipped with overhead projectors, chalk boards, and video feedback arrangements. This block also has two discussion rooms where group exercises and discussion can be conducted for a batch of up to 30 participants each.

3.1.2 Faculty cum Laboratory Block

i) *Organizational Behaviour Laboratories*

Two laboratories have been established for conducting practical training in the area of organizational behaviour (OB) and human resources management. The large group process room can accommodate about 30 trainees, while the small group process room can cater to the needs of 15 trainees. These laboratories are equipped with chalk boards and overhead projectors. They are furnished with specially designed furniture which can be rearranged keeping in view the requirements of the different exercises. Individual and group exercises in team building, leadership, and group dynamics are conducted in these laboratories. These are done mainly through structured experiences for training in human behaviour. These

laboratories are in close proximity of the video laboratory, which facilitates the wheeling in of video equipment when required for the OB exercises.

ii) Video Instruction and Production Laboratory

This laboratory comprises two television studios and a master control room. It is used for intensive training of scientists in the preparation of effective video programmes in agriculture. The laboratory is also used for producing educational videos. In addition, this laboratory helps and collaborates with the various units of the Academy to provide video feedback in several programmes on certain specific exercises like role play, communication skills and group activities.

iii) Audio Visual Laboratory

This laboratory is equipped with all the modern audio visual equipments including overhead projector colour transparency making machines; it also provides training to the scientists in the use of the hardware as well as in the software technology related to various audio visual aids.

iv) Self Aided Instructional Laboratory

This laboratory provides an opportunity for the trainees to learn and acquire skills in developing and using various structured and programmed instructional lessons. A set of six carrels, each with a caramate for playing a tape slide programme, a video cassette player with monitor for a video tutorial, and a computer for a programmed instruction are available in this laboratory.

v) Computer Lab

A miniframe computer WIPRO S-386 with 4 MB RAM and 170 MB hard drive along with 12 terminals cater to the computing needs of the Academy. Required peripherals such as laserjet printer, dot matrix printers, near letter quality printers and a plotter are also available in the Lab. Further, a few micros have been procured additionally to train the trainee scientists effectively.

These computers are being used in the preparation of paybills, accounting and budgeting, recruitment processing, secretarial work and also in preparation of databases. The scientist trainees, faculty and staff are trained in the use of various packages such as Word Perfect, Lotus and dBase. This laboratory has been identified as the training facility by the National Agricultural Research Project.



Computer learning in progress

A Desktop Publishing System of Apple Macintosh-SE and Laserwriter II is also in use with the required software. These are useful in bringing out quality publications by the offset press of the Academy.

vi) *User Instruction and Information Resource Lab*

This laboratory provides training to the users in the access of information. A CD-ROM work station with a DCM TURBO 2000 PC-AT has been established in this laboratory for this purpose.

3.1.3 Library

The Academy has established a specialized and a well organized library to meet the information needs of the faculty members, staff, and visiting scientists/trainees. Specialized collection focuses on agricultural research management, science policy, systems studies, project management, futurology, forecasting, educational technology, teaching, transfer systems/policies, women studies, information management, research communication, computer/information technology, human resource development, administration, etc. International corner, foreign tour reports section, and training education abroad information cell are the unique sections of this library. About 400 titles were added during the year. The library now holds about 20,000 books and subscribes to 425 journals/serials.

During February 1993, the library moved to its spacious building, wherein database services, information resource laboratory, learning resource laboratory, micrographic, reprographic, and video library have been established. To facilitate information storage and retrieval, the computer software CDS/ISIS 3.0 is in operation.

The General Information Service Terminal of National Informatics Centre (GISTNIC) installed in the library makes available basic information about India, important national and international information, agricultural/census/economic statistics, railway/air time-table, recreational and tourist information. GISTNIC terminal at the Academy is extensively used for supplying statistical information to the scientist trainees, faculty and staff of the Academy. A secondary publication titled Agricultural Research Management Abstract (ARMA) is brought out by the Information and Documentation Unit.

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3.1.4 Conference Facilities

The Academy has excellent modern conference facilities. This consists of three halls, which are the conference hall, the committee room and the faculty seminar hall. The conference hall is a tastefully decorated air-conditioned facility for conferencing and can accommodate about 125 persons. The hall is fitted with a modern conferencing public address system. The air-conditioned committee rooms has an artfully designed interior, and provides meeting facilities for about 50 people. The faculty seminar hall is also an air-conditioned facility and can accommodate about 30 people. There is a lounge of matching capacity adjacent to this seminar hall. All these rooms are equipped with different kinds of projection facilities.

3.1.5 Residential Facilities

The Academy has a total of 95 residential quarters in different categories for the purpose of accommodating the various scientists, technicians, and staff.

3.2 Services

3.2.1 Administration, Finance and Accounts

This wing of the Academy is organized into five sections: Administration I deals with establishment matters; Administration II with cash and bills; Administration III with stores and purchases; Administration IV with estate management and training; all matters pertaining to expenditure and bank reconciliation statements, monthly and annual accounts, budget and revised estimates, internal audit etc are dealt by the audit and accounts section.

This wing provides administrative, financial, and accounts support to all the units of the Academy. Further, the Chief Administrative Officer and the Finance and Accounts Officer function as regular faculty members in delivering lectures on various rules and regulations of ICAR to ARS scientists and other trainees.

3.2.2 Production and Publication Services

This cell undertakes publishing and printing of all the materials brought out by the Academy. This cell is equipped with a Desk Top Publishing unit from Apple Macintosh, offset printing press, swift 300 deluxe, plate making and binding systems. The cell also has a screen printing facility.

3.2.3 Farm Services

A congenial atmosphere for academic activities is provided by developing an aesthetic and environmentally compatible surroundings to the Academy. The farm services is involved in maintaining this environment by establishing lawns, flowering annuals, borders, hedges, shrubs, and trees in and around the Administrative Block, Academic Block, Halls of Residence, and Scientist Home.

3.2.4 Hostel Services

All the training programmes conducted by the Academy are residential in nature and the institution has a Halls of Residence with 150 fully furnished single seated rooms with attached bath; excellent dining and recreational facilities are also available in this hall. A Scientist Home with 32 double suites is also available for the use of senior officials and scientists.

3.2.5 Hindi Cell

The Hindi Cell of the Academy discharges the constitutional obligation of implementing the official language policy and rules and regulations of the Govt. of India, issued from time to time. The Cell also brings out publications like NAARM Bulletin, NAARM News and Annual Report in Hindi.

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4. TECHNICAL INFORMATION



4.1 Human and financial resources

HUMAN RESOURCES			
Sl. No.	Name of the post/ Category	Sanctioned strength	Posts filled as on 31-03-93
1.	Scientific	43	19
2.	Technical	41	41
3.	Administrative	49	48
4.	Auxiliary	28	26
5.	Supporting	37	37
Total		198	171

FINANCIAL RESOURCES

Head of Account	Actuals during 1992-93			Revised Estimates 1993-94		
	Non-Plan	Plan	Total	Non-Plan	Plan	Total
Pay & Allowances	6,784,400	3,326,800	10,111,200	8,521,000	6,000,000	14,521,000
Travelling Expenses	139,100	649,600	788,700	160,000	500,000	660,000
Recurring Contingencies	3,820,500	890,200	4,710,700	519,000	1,000,000	1,519,000
Non-Recurring Contingencies						
1. Equipments	--	2,995,400	2,995,400	--	--	--
2. Works	--	932,300	932,300	--	500,000	500,000
3. Others	--	--	--	--	--	--
Total	10,744,000	8,794,300	19,538,300	9,200,000	8,000,000	17,200,000

4.2 New Resource Materials Developed for Training/ Teaching

Estimates 1993-94	
Plan	Total
6,000,000	14,521,000
500,000	660,000
1,000,000	1,519,000
--	--
500,000	500,000
--	--
8,000,000	17,200,000

Anwer, M.M.	A scheme for training in research management
Balaguru, T.	Methods for popularizing fodder crops
	On-farm testing methodologies
	Farming Systems Approach
Balaguru T. and Manikandan, P.	Case exercise for group task accomplishment
	Bibliography on training and its management
Balaguru, T. Manikandan, P. and Kalla, J. C.	MIS case studies for two regional research stations of an SAU
Challa, J. and Anwer M.M.,	Personality types
Challa, J., Anwer, M.M. and Suresh Kumar, M.	Delegation process
Challa J. and Kalla, J. C.	Intellectual property rights vis-a-vis Indian agriculture : A view point
Chelvan, R.C.	Communication technologies for the future
Chelvan, R.C. and Anwer, M.M.	Use of video in horticulture education, extension, and practice

- Gaddagimath, R.B. Computer applications for ISR in agricultural libraries and information centres
- Communication format (CCF)
- Gopalam, A. Self-instructional systems: approach to instruction in higher education for agriculture
- Introduction to ADII model
- Computer aided instruction
- Kumar, P. Contingency planning in agriculture
- Dimensions of forest policy
- Muralidhar, U. Basics of computers
- Word processing - An overview
- Office automation
- Virus and vaccines
- A tutorial on dbase-III
- Exercises on MSTAT-C
- Exercises on MICROSTAT
- Exercises on Word Perfect
- Exercises on Lotus
- Computer applications in agriculture

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mat (CCF)	Rao, V.R.	Research planning Research project Research project proposal
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An overview		Organising university information service
	Reddy, K.M.	HyperCard stacks for demonstra- tion of 'Animation'
-III		HyperCard stacks on 'Practices in Dairying'
AT-C	Sabarathnam, VE.	Process of scientific manpower planning in agricultural research system Manpower planning for agricultural research and management of tech- nology Methods and techniques of forecast- ing manpower requirement in agri- cultural research system
ROSTAT		
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Saha, S.N.

Methods and techniques of supply forecasting of scientific manpower

Self-learning exercise in demand forecasting for scientific manpower

Group exercise for manning norms for workload estimation of technology development projects

Self-learning exercise for supply forecasting of scientific manpower

Promoters and retarders affecting efficiency of manpower

Self-learning exercise on estimation of manpower efficiency

Management of manpower time

Self-learning exercise on time management of scientific manpower

Problems and issues of forecasting scientific manpower requirement

Problems and issues in allocation of technical, administrative, and supportive manpower for scientific manpower

Issues on principles of time management of scientific manpower agricultural research management process

Why manage agricultural research

Production potential and constraints in forestry

ues of supply fic manpower		Systems approach in project devel- opment
e in demand tific manpower		Brainstorming- a tool in training
nanning norms tion of technol- objects	Samanta, R.K.	Brainstorming and synetics
se for supply tific manpower		Communication skill exercises
ers affecting wer		Management of instructional aids for training excellence
se on estimation ncy		Human communication in organiza- tion for change and development
power time		Organizational communication
se on time man- c manpower		Organizational renewal and change process in organization
s of forecasting r requirement		Follow your instructions - instruc- tional communication
s in allocation of ative, and sup or scientific	Suresh Kumar, M.	Values worksheet - perception
of time man- c manpower n management		Notes on writing of confidential character rolls
ltural research		General service conditions - an overview
l and constraints	Venkateswarlu, K.	Case studies on management of agricultural research stations - operations management

4.3 Publications

4.3.1 Books

1. Raman, K.V. and Balaguru, T. (Eds.) 1992. Farming Systems Research in India : Strategies for Implementation. Proceedings of the National Workshop held at NAARM, Hyderabad from 25th to 28th November, 1991. National Academy of Agricultural Research Management, Hyderabad in association with Ford Foundation, New Delhi.
2. Sabarathnam, V.E. 1993. Planning and Utilization of Scientific Manpower in Indian Agricultural Research. National Academy of Agricultural Research Management, Hyderabad, 120 p.

4.3.2 Manuals

1. Rao, K.P.C., Gadewar, A.U. and Kalla, Jagdeesh C. 1992. Field Experience Training Manual. National Academy of Agricultural Research Management, Hyderabad.
2. Gaddagimath, R.B., Sivaprasad, Y.V. and Shanmuganathan, T. 1992. Working manual for UNESCO's mini-micro CDS/ISIS version 2.32. National Academy of Agricultural Research Management, Hyderabad.

4.3.3 Research Papers and Popular Articles

1. Balaguru, T. 1992. Farming systems research and extension (FSR/E) : An innovative approach to technology generation and transfer in agriculture. Eastern India Farming Systems Research and Extension Newsletter, VI (4) : 10-19.
2. Balaguru, T. and Kalla, Jagdeesh C. 1992. Research management training of agricultural scientists in India. In : Singh, U.K. and Trivedi, P.R (Eds.). HRD for production, Prospects and constraints. Proceedings of the 8th All India Management Congress, New Delhi, October 18-19, 1992 pp: 143-52.

3. Gaddagimath, R.B. 1993. User instruction programme in academic libraries. In : Current Studies in Library and Information Science, edited by Karasiddappa, C.R. S.L.Sangam and B.S. Maheswarappa , Manak Publishers, New Delhi pp: 292-303.
4. Muthuraman, P. and Gadewar, A.U. 1993. Participatory technology generation. Kurukshetra, XLI.
5. Narsimha Rao, G., Samanta, R.K., Reddy K.M. and Kishore, D. 1992. A.V.Aids for better learning. The Hindu, October 5, 1992.
6. Sabarathnam, V.E. 1992. Interface of rural development officials with non-officials. Kurukshetra, XL (12), 34-40.
7. Samanta, R.K. 1992. Participatory rural appraisal for purposive development. Kurukshetra, XL (11): 31-3
8. Samanta, R.K. 1992. Organization renewal and transfer of technology in agriculture. Agricultural Situation in India, XLVII (11):403-6.
9. Samanta, R.K. 1992. Participatory message construction - A new vistas in development communication research. Communicator, 27(4):3-6,13.
10. Samanta, R.K. 1992. Brainstorming - the new method. The Hindu, January 7, 1993.
11. Samanta, R.K. and Kalla, Jagdeesh C. 1992. The Buzz word for healthy interaction. The Hindu, December 10, 1992.
12. Samanta, R.K. and Kalla, Jagdeesh C. 1992. Organizational renewal - key for development. The Hindu, February 18, 1993.
13. Samanta, R.K. and Reddy, M.V. 1992. Perception of agricultural research service scientist probationers about their services. Manpower Journal, 27(4):61-8.

4.4 Faculty Development

Participation and Presentation of Papers in Seminars, Conferences, and Workshops

1. Dr.M.M.Anwer participated in the workshop on "The use of national language in Indian agricultural research institutions" held at NAARM from December 15-19,1992.
2. Dr.T.Balaguru participated in the 8th All India Management Congress on HRD for productivity held at New Delhi from October 18-19,1992, and presented a paper on "Research management of agricultural scientists in India"
3. Dr.T.Balaguru participated in the Workshop on "Technology generation and transfer for agricultural development" held at NAARM from November 24-27,1992 and presented a paper on "Farming Systems Research and Extension (FSR/E)".
4. Mr.R.B.Gaddagimath presented a paper on "Resource sharing in libraries" at National Workshop and Summer Institute for Librarians, Karnatak University, Dharwad, held in June,1992.
5. Mr.R.B.Gaddagimath presented a paper on "User instruction in libraries : impact of Dr.S.R.Ranganathan's five laws" at the Seminar on Dr.S.R.Ranganathan's Centenary Celebrations at Osmania University, Hyderabad, held in August 1992. The paper was jointly authored by Mr.R.B.Gaddagimath and Mr.T.Shanmuganathan.
6. Mr.R.B. Gaddagimath participated in the International Conference of IFLA from September 1-4,1992 at New Delhi.
7. Mr.R.B. Gaddagimath, presented a paper on, "Automation of CDS/ISIS package at NAARM for user friendly usage" at the National Seminar on Library Automation in R & D organizations held at Defence Metallurgical Research Laboratories, Hyderabad, in September,1992. The paper was jointly authored by Mr.R.B.Gaddagimath, Mr.T.Shanmuganathan and Mr.Y.V.Sivaprasad.

in Seminars,

Workshop on "The
research insti-
-19,1992.

All India Man-
held at New Delhi
paper on "Re-
s in India"

Workshop on "Tech-
cal development"
2 and presented a
Extension (FSR/

er on "Resource
op and Summer
, Dharwad, held

per on "User in-
nganathan's five
than's Centenary
ad, held in August
B.B.Gaddagimath

the International
1992 at New Delhi.

per on, "Automa-
er friendly usage"
mation in R & D
al Research Labo-
l. The paper was
B.B.Gaddagimath,
rasad.

8. Mr.R.B.Gaddagimath participated in the National Users Meet of CDS/ISIS at National Chemical Lab, Pune, organized by DSIR, New Delhi from January 6-8,1993.

9. Dr.A.U.Gadewar presented a paper, on "An experimental design to study effectiveness of communication technology" in the workshop on "Communication Research Methodology" held at NAARM from June 22-30,1992. This paper was jointly authored by Dr. A.U.Gadewar, Dr. Devesh Kishore and Dr.Sreenath Dixit.

10. Dr.A.U.Gadewar presented a paper on "Communication with extension clients" in the Workshop on "Meaningful learning as a communication process" held at NAARM from July 7-10,1992.

11. Dr. Sreenath Dixit and Dr. A.U.Gadewar presented a paper on "Action research in communication" in the workshop on "Communication research methodology" held at NAARM from June 22-30,1992.

12. Dr.A.U.Gadewar participated in the Workshop on "Participatory methods in extension and group management skills" held at National Institute of Agricultural Extension Management, Hyderabad from February 8-11,1993.

13. Dr.A.Gopalam participated in the Workshop for officers of the Official Language Implementation Committee. This was organized by Rajbhasha Samsthan at New Delhi on 19th and 20th October,1992.

14. Dr.Jagannadham Challa participated in the National Workshop on "Information systems for livestock research and development" held at Central Sheep and Wool Research Institute, Avikanagar, Rajasthan from February 1-3,1993.

15. Dr.Jagdeesh C.Kalla participated in the Workshop on Implementation of Land Resources in A.P., organized at National Institute of Rural Development from May 15-17,1992.

16. Dr.Jagdeesh C.Kalla and Dr.A.Gopalam participated in the Second Academic Conference of AMDISA held at Nepal Administrative Staff College, Kathmandu, from May 2 - 7,1992. Dr.Kalla presented a paper on "NAARM : A profile in dynamics of agricultural management institution". Dr.Gopalam presented a paper on "Application of expert system for the evaluation of research projects in ICAR".

17. Dr.P.Kumar participated as a resource person in Workshop on "Planning and utilisation of scientific manpower for agricultural research systems" organized at NAARM from October 12-16,1992.

18. Dr.P.Kumar participated as a resource person in the Workshop on "Vocationalisation of education and reorganization of agricultural courses into modules" held at NAARM from January 5-9,1993.

19. Mr.U.Muralidhar participated in Indian Computer Congress held at New Delhi from December 6-8,1992.

20. Dr.K.P.C.Rao presented a paper "Is agricultural policy a panacea?" at the National Seminar on "Agricultural policy and intellectual property rights in agriculture" organized by ARS Scientists Forum at IARI, New Delhi from November 16-18,1992. This paper was jointly authored by Dr.K.P.C.Rao and Dr.Jagdeesh C.Kalla.

21. Dr.K.P.C.Rao presented a paper on "New economic policies and rural development" at the National Seminar on "Planning and rural development" organized at Nagarjuna University, Guntur from March 3-4,1993.

22. Dr.S.Shanmugam participated in the Workshop on "Management of agricultural research stations : operations management" held at NAARM from September 22-25,1992.

4.5 Visitors

* Dr. Olu Makinde, Director and Dr. Bimbo Adewuni, Deputy Director, Agriculture and Rural Management Training and Institute, Nigeria visited on April 10, 1992.

* Dr. Shawki Barghouti, Chief Agricultural Operations for India Department, World Bank, Washington DC and Dr. Michael Macklin, Acting Chief, Agriculture Unit, World Bank, New Delhi visited on May 20, 1992.

* Fifteen students of third year B.Sc. (Hort.) from Dr. Y.S. Parmar University of Horticulture, Solan, Himachal Pradesh visited on June 18, 1992.

* Dr. Selina Ahsan, Principal Scientific Officer, Bangladesh Agricultural Research Council, Dhaka visited on July 1, 1992.



The World Bank Team with the Director in the Library

- * Seventy final year students of B.Sc.(Agriculture) from Agricultural College and Research Institute, Killikulam, Tamilnadu visited on July 13,1992.
- * Thirty seven students of final year B.Sc.(Ag.) from Pandit Jawaharlal Nehru College of Agriculture, Karaikal, Pondy, visited on July 29,1992.
- * Mr. Anne Waldrop, Programme Officer UNDP, New Delhi visited the Academy on October 9,1992.
- * Mr. Yuan Longping, Director of Human Hybrid Rice Research Centre, Changsha, Hunan, China, and Mr.Chang Xiang Mao, Deputy Director of the Centre visited on October 31,1992.
- * A delegation from Nepal, comprising Mr.Yadav Dev Pant, Director, Department of Agricultural Development, Nepal, Mr.Satyendra P.Singh, Chief Regional Agricultural Training Centre, Nepal gunj, and Mr.Saligram Adhikary, Chief, Regional Agricultural Training Centre, Bhairawa, visited the Academy on November 13,1992.
- * Mr. Zhu Faughma, Lecturer, Department of Agronomy, South China Agricultural University, visited the Academy on February 24,1993 under Indo-China cultural exchange programme.
- * A group of six scientists and farmers from Rajasthan led by Dr.Purohit of CAZRI, Jodhpur visited the Academy to know about the activities.
- * Mr.John Canziv, an agricultural expert from Canada visited the Academy on March 1,1993.

4.6 NAARM Alumni Association

The total membership of the Association stands at 473 as on March 31, 1993. The following are the office-bearers of the Association for 1992-93.

PATRON	:	Prof. V.L. Chopra, DG, ICAR
PRESIDENT	:	Dr. Jagdeesh C. Kalla Director, NAARM
GENERAL SECRETARY	:	R. Chithiraichelvan Scientist (SG), NAARM
VICE -PRESIDENTS (2)		
Local	:	Dr. Vilas A. Tonapi NRCS, Hyderabad
All India	:	Dr. A.K. Saxena CIAE, Bhopal
JOINT SECRETARIES (4)		
Eastern Region	:	Dr. K.K. Sharma ILRI, Ranchi
Northern Region	:	Dr. Sanjeev Gupta IGFRI, Jhansi
Western Region	:	Mr.A. Chaturvedi NBSS LUP, Nagpur
Central Region	:	Mr.U.R. Badegaonkar CIAE, Bhopal
TREASURER	:	Mr. B. Pandu Reddy Finance & Accounts Officer, NAARM

4.7 Committees

4.7.1 Management Committee

The Academy is guided in planning, organizing, and conducting various types of training programmes and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation in terms of physical targets and schedules. The Management Committee also helps in formulation and consolidation of proposals for Five Year Plan and Annual Plan and takes up periodical review of progress of development schemes. The Management Committee assists the Director of the Academy in discharging his functions; it also focuses its attention on the research and other programmes of the Academy, and ensures their implementation. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests remedial measures.

The 20th meeting of the Management Committee of the Academy was held on March 23, 1993. The composition of the Management Committee of NAARM is as follows :

- | | | |
|----|--|----------|
| 1. | Dr.Jagdeesh C.Kalla
Director, NAARM
Rajendranagar, Hyderabad | Chairman |
| 2. | Dr.P.N.Mathur
ADG (Extension)
ICAR, New Delhi | Member |
| 3. | Dr.M.S.Chari
Director, CTRI
Rajahmundry, Andhra Pradesh | Member |
| 4. | Dr.Punjab Singh
Director, IGFRI
Jhansi | Member |

- | | | |
|----|--|------------------|
| 5. | Dr. A.Ahmed
DDG (Education), ICAR
New Delhi | Member |
| 6. | Mr.B.Pandu Reddy
Finance & Accounts Officer
NAARM
Rajendranagar
Hyderabad | Member |
| 7. | Mr.M.Suresh Kumar
Chief Administrative Officer
NAARM
Rajendranagar
Hyderabad | Member Secretary |

4.7.2 Steering Committee on Advanced Centre on Agricultural Education and Research Management

This Committee was initially called the Sub-project Advisory Committee and has been later renamed as Steering Committee. The 8th meeting of the Steering Committee on the Advanced Centre on Agricultural Education and Research Management was held on 4th April, 1992 at NAARM, Hyderabad. The Committee reviewed the progress of the project and provided broad guidelines for its implementation based on the suggestions of user agencies. The following are the members

- | | | |
|----|--|----------|
| 1. | Dr.Jagdeesh C. Kalla
Director, NAARM, Hyderabad | Chairman |
| 2. | Dr.Harswaroop Singh
Lt. Governor, Pondicherry | Member |
| 3. | Dr.K.V.Raman
Ex-Director, NAARM, Hyderabad | Member |

- | | | |
|-----|---|--------|
| 4. | Dr.M.V.Rao
Vice-Chancellor, APAU, Hyderabad | Member |
| 5. | Dr. Gnana Prakasam
Vice-Chancellor, TN University
of Veterinary and Animal Sciences
Madras | Member |
| 6. | Dr.A.A.Ahmed
D.D.G(Education), ICAR, New Delhi | Member |
| 7. | Dr.S.C.Sarma
Director, NDRI, Karnal | Member |
| 8. | Dr.P.B. Mathur
A.D.G. (UNDP)-& NDP, ICAR
New Delhi | Member |
| 9. | Dr.P.Joshi
Director of Research, Rajasthan
Agricultural University, Bikaner
Rajasthan | Member |
| 10. | Dr.A.P.Vyas
Director of Research, Gujarat
Agricultural University, Ahmedabad | Member |
| 11. | Dr.M.V.R.Prasad
Director, Directorate of Oilseeds
Research, Rajendranagar, Hyderabad | Member |
| 12. | Dr. Raghavendra Rao
Professor
NAARM, Hyderabad | Member |
| 13. | Dr.I.J.Singh
Professor, Department of
Agricultural Economics, H.A.U.
Hissar, Haryana | Member |

Member

14. Mr.M.Suresh Kumar
Chief Administrative Officer,
NAARM, Hyderabad
- Member Secretary

Member

15. Dr. S.I.Zakhariev
FAO Representative in India & Bhutan
FAO of the United Nations
55 Max Muller Marg (ex-Lodi Estate)
New Delhi
- Ex-officio
Member

Member

4.7.3 Academic Council

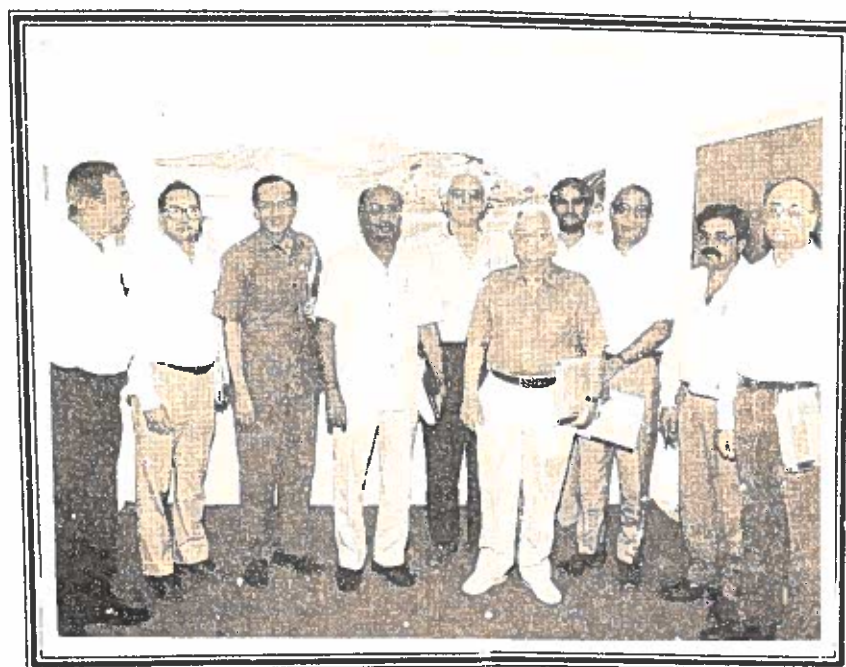
Member

1. Dr.Jagdeesh C.Kalla
Director, NAARM
- Chairman

Member

2. Dr.P.Kumar
Incharge, Agricultural Research
Systems & Policies Unit
- Member

Member



Member

Member

Member

Member

Members of the Steering Committee

- | | | |
|----|---|------------------|
| 3. | Dr.R.K.Samanta
Incharge, Transfer of Technology
Systems & Policies Unit | Member |
| 4. | Dr.A.Gopalam
Incharge, Educational Systems &
Technology Unit | Member |
| 5. | Dr.M.M.Anwer
Incharge, Human Resources
Development Unit | Member |
| 6. | Mr.R.B.Gaddagimath
Incharge, Information &
Documentation Unit | Member |
| 7. | Dr.V.Raghavendra Rao
Incharge,
Project Management Unit | Member Secretary |

4.7.4. Research Council

- | | | |
|----|---|----------|
| 1. | Dr.Jagdeesh C.Kalla
Director, NAARM | Chairman |
| 2. | Dr.V.Raghavendra Rao
Incharge, Project Management Unit | Member |
| 3. | Dr.P.Kumar
Incharge, Agricultural Research
Systems & Policies Unit | Member |
| 4. | Dr.R.K.Samanta
Incharge, Transfer of Technology
Systems & Policies Unit | Member |
| 5. | Dr.A.Gopalam
Incharge, Educational Systems &
Technology Unit | Member |

Member	6. Dr.M.M.Anwer Incharge, Human Resources Development Unit	Member
Member	7. Mr.R.B.Gaddagimath Incharge, Information & Documentation Unit	Member
Member	8. Dr T.Balaguru Faculty Member, ARSP Unit	Member Secretary
Member		

4.7.5 Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and rules and regulations of Government of India that are issued from time to time. The Committee met on June 19, 1992, October 30, 1992, December 31, 1992, and March 26, 1993, and discussed several issues relating to the implementation of Hindi language. The following were the members :

Chairman	1. Dr.Jagdeesh C.Kalla, Director	Chairman
Member	2. Dr.A. Gopalam, Senior Scientist	Vice-Chairman
Member	3. Dr.K.P.C.Rao, Senior Scientist	Member
	4. Dr.D.Rama Rao, Senior Scientist	Member
Member	5. Mr.M.Suresh Kumar, Chief Administrative Officer	Member
Member	6. Mr.C.P.Mohan Kumar Finance & Accounts Officer	Member

7. Mr. K.R.Prabhakar, Programme Officer (CCTV)	Member
8. Mr. U.Muralidhar Programmer (Computer)	Member
9. Mr. R.V.V.S.Prakasa Rao, Editor	Member
10.Mr. Abhijit Bhattacharjee, PRO	Member
11.Ms. Jhansi Laxmi, Sr.P.A	Member
12.Mr. D.Venkateswarlu Technical Officer (Hindi)	Member Secretary

Member

4.8 Personnel

Member

1. Dr. Jagdeesh C.Kalla, Director

Director's Cell

Member

2. Ms. L.Jhansi Lakshmi, Senior Personal Asst
3. Mr. T.V.Rama Das, Junior Stenographer
4. Mr. P.Swamy, SSG-II

Member

Agricultural Research Systems & Policies Unit

Member

5. Dr. P.Kumar, Principal Scientist
6. Dr. VE.Sabarathnam, Senior Scientist
7. Dr. T.Balaguru, Senior Scientist
8. Dr. K.P.C.Rao, Senior Scientist
9. Dr. K.Vidyasagar Rao, Senior Scientist
10. Ms. A.Vijaya Lakshmi, Junior Stenographer
11. Ms. A.Mercy, Junior Stenographer
12. Mr. M.Shyam Rao, SSG-I

Member
Secretary

Educational Systems and Technology Unit

13. Dr. Devesh Kishore, Principal Scientist
14. Dr. A.Gopalam, Senior Scientist
15. Dr. K.M.Reddy, Senior Scientist
16. Dr. D.Rama Rao, Senior Scientist
17. Mr. N.R.Nageswara Rao, Technical Officer (T-5)
18. Mr. M.Venkatesh, Stenographer
19. Mr. M.Yadaiah, SSG-II

Transfer of Technology Systems & Policies Unit

20. Dr. R.K.Samanta, Senior Scientist
21. Mr. R.Chithiraichelvan, Scientist (SG)
22. Dr. A.U.Gadewar, Senior Scientist
23. Mr. L.Venkateswarlu, Photographer (T-II-3)
24. Mr. Bansidhar Nayak, Projectionist (T-II-3)
25. Mr. D.Rajagopal Rao, PAE Operator
26. Mr. S.Sunder Raj, Media Operator (T-1)
27. Mr. P.Anand Kumar, Junior Stenographer
28. Mr. G.V.Bikshapathi, SSG-II

Project Management Unit

- 29. Dr.V.Raghavendra Rao, Principal Scientist
- 30. Dr.S.N.Saha, Principal Scientist
- 31. Dr. K.Venkateswarlu, Senior Scientist
- 32. Dr. P.Manikandan, Senior Scientist
- 33. Ms. Sarada Samanta, Senior Stenographer
- 34. Ms. T.Vanisri, Junior Stenographer
- 35. Mr. K.Satyanarayana, SSG-II

Human Resources Development Unit :

- 36. Dr. M.M.Anwer, Senior Scientist
- 37. Dr. Jagannadham Challa, Senior Scientist
- 38. Dr. S.Shanmugam, Senior Scientist
- 39. Ms. S.Sesha Sai, Junior Stenographer
- 40. Mr. M.Narsinga Rao, SSG-I

Information and Documentation Unit

- 41. Mr. R.B.Gaddagimath, Librarian (T-8)
- 42. Ms. P.V.Nirmala, Cataloguer (T-4)
- 43. Mr. Y.V.Siva Prasad, Documentation Asst (T-II-3)
- 44. Mr. T.Shanmuganathan, Library Assistant (T-2)
- 45. Ms. S.Shanti, Junior Stenographer
- 46. Ms. D.Dhanalakshmi, Junior Clerk
- 47. Mr. Venkatesham, Xerox Machine Operator
- 48. Mr. M.Ashok, SSG-II

Hindi Section

- 49. Mr. D.Venkateswarlu, Technical Officer (T-5)
- 50. Ms. J.Renuka, Hindi Assistant
- 51. Mr. S.Pradeep Singh, Hindi Translator
- 52. Ms. P.Raja Rajeswari Jyothi, Junior Clerk

Computer Laboratory

- 53. Mr. U.Muralidhar, Programmer (Computers) T-7
- 54. Mr. M.J.Pradeep Kumar, Computer Operator (T-II-3)
- 55. Mr. K.Venkateswara Kumar, Statistical Asst (T-II-3)
- 56. Mr. P.Mohan Singh, Computer Assistant (T-II-3)

Video Instruction and Production Laboratory

- 57. Mr. K.R.Prabhakar, Programme Officer (CCTV) T-7
- 58. Mr. Ch.Janardhan Rao, Video Cameraman (T-4)
- 59. Mr. N.Naresh Kumar, VCR-Tele Cine Operator (T-2)
- 60. Mr. C.Bikshapathi, SSG-II

Production and Publication

61. Mr. R.V.V.S.Prakasa Rao, Editor (T-6)
62. Mr. N.Narasimha Reddy, Offset Machine Operator
63. Mr. Y.Vijay Kumar, Assistant Editor (T-II-3)
64. Mr. K.Anil Kumar, Reprographic Assistant (T-II-3)
65. Mr. M.Shekhar Reddy, Dark Room Assistant (T-2)
66. Mr. S.Swamy, Assistant Gestetner Operator
67. Mr. Prashant Gaikwad, Binder
68. Mr. C.Chandramouli, SSG-I

Farm Section

69. Mr. V.Murali, Farm Superintendent (T-6)
70. Mr. K.Rangasamy, Field Assistant (T-4)
71. Mr. M.A. Basith, Lab Technician (T-4)
72. Mr. B.Veeraiah, Farm Assistant (T-I-3)
73. Mr. Ahire Laxman Mahar, Farm Assistant (T-1)
74. Mr. K.Shivaiah, Technician Electrical (T-1)
75. Mr. P.Bhaktavatsalam, Junior Clerk
76. Mr. M.Padmaiah, Tractor Driver
77. Mr. P.Iswari, Power Tiller Driver
78. Mr. P.Balraj, SSG-II
79. Ms. N.Sukunamma, SSG-I
80. Mr. S.Jangaiah, SSG-II
81. Ms. B.Bharatamma, SSG-I
82. Mr. G.Pentaiah, SSG-I
83. Mr. R.Sattaiah, SSG-I
84. Ms. G.Anasuya, SSG-I
85. Mr. K.Hemlal, SSG-I

Hostel Services

86. Mr. Zameer Ahmed, Manager (Hostel Services)
87. Mr. Sham Bahadur, Catering-in-charge
88. Ms. Savithri Srinivasan, Mess Supervisor
89. Mr. C.J.Samuel, Junior Clerk
90. Ms. D.Sangamma, Cook
91. Mr. Khalid, SSG-II
92. Ms. B.Santhamma, SSG-II
93. Mr. B.Subba Rao, SSG-II
94. Ms. S.Shakuntala, SSG-II
95. Mr. A.Yadaiah, SSG-I

96. Mr. M.Krishnaiah, SSG-I
97. Mr. C.Anjaiah, SSG-I
98. Ms. K.Kalavathi, SSG-II
99. Ms. M.Yadamma, SSG-I
100. Ms. I.Bharathamma, SSG-I
101. Ms. T.Jangamma, SSG-I
102. Ms. Nazia Parveen, SSG-I

Administration

103. Mr. M.Suresh Kumar, Chief Administrative Officer
104. Mr. Mohammed Habeebuddin, Assistant
Administrative Officer
105. Mr. K.Janaki Ramaiah, Superintendent
106. Mr. T.Ashok Kumar, Superintendent
107. Mr. Y.Sankara Rao, Assistant
108. Mr. P.P.Brahmaji, Assistant
109. Mr. K.S.Sharma, Assistant
110. Mr. N.Raghunath, Stenographer
111. Mr. P.Neelakantam, Assistant
112. Mr. P.G.Kohad, Assistant
113. Mr. P.Anantha Reddy, Technician (Instruments)
114. Mr. K.Prabhudas, Senior Clerk
115. Mr. M.Narasimha Rao, Senior Clerk
116. Mr. T.Srinivas, Senior Clerk
117. Mr. M.Dinesh, Senior Clerk
118. Ms. K.Swarajya Laxmi, Junior Library Assistant
119. Mr. A.M.Venkateswarlu, Compounder
120. Mr. G.Raj Reddy, Junior Clerk
121. Ms. B.Padma Saroja, Junior Clerk
122. Mr. P.Srinivasu, Junior Clerk
123. Mr. K.Srinivasa Rao, Junior Clerk
124. Mr. C.Phani Raj, Junior Clerk
125. Ms. P.D.Jessie Ecclicia, PMBX Operator cum
Receptionist
126. Mr. M.Sridhar, Junior Clerk
127. Mr. P.Venkatesh, Junior Clerk
128. Mr. K.R.Ghanshyam, Junior Clerk
129. Mr. B.H.Dharma Raj, SSG-III
130. Mr. J.Chandraiah, SSG-I

Typing Pool

- 131. Ms. V.Shailaja, Junior Stenographer
- 132. Ms. K.K.Rukmani Ammal, Junior Stenographer

Public Relations Cell

- 133. Mr. Abhijit Bhattacharjee, Public Relations Officer
- 134. Mr. A.Murali Mohan Rao, Artist
- 135. Mr. P.Namdev, Graphic cum Visualiser
- 136. Mr. P.Vijender Reddy, Museum Assistant
- 137. Ms. Y.Anuradha, Junior Stenographer

Civil Engineering Cell

- 138. Mr. Sohail Ahmad Khan, Junior Engineer (Civil)
- 139. Mr. K.Veeranarsaiah, Carpenter
- 140. Mr. N.Prabhakar Rao, Plumber

Audit and Accounts Section

- 141. Mr. B.Pandu Reddy, Finance & Accounts Officer
- 142. Ms. M.Girija Kumari, Junior Stenographer
- 143. Mr. C.Bagaiah, Assistant
- 144. Mr. S.Balakamesh, Junior Clerk
- 145. Mr. R.Chandra Babu, Junior Clerk
- 146. Ms. K.Radha Sujatha, Junior Stenographer
- 147. Ms. Y.Gayathri, Junior Clerk
- 148. Ms. G.Manibai, SSG-III

Transport Section

- 149. Mr. K.Sattaiah, Driver
- 150. Mr. S.K.Peera, Driver
- 151. Mr. M.M.Mohan Rao, Vehicle Mechanic (T-2)
- 152. Mr. G.Muthyalu, Driver
- 153. Mr. U.V.Ratnam, Driver
- 154. Mr. N.Ashok, Driver
- 155. Mr. T.Laxman, Driver
- 156. Mr. B.Narayana, SSG-III

Security Wing

- 157. Mr. B.Ch.Satyanarayana, Security Officer cum Caretaker
- 158. Mr. B.K.Venkatram, Pump Driver
- 159. Mr. M.Srinivasa Rao, Pump Driver
- 160. Mr. Phool Kumar, SSG-II
- 161. Mr. L.Satyanarayana, SSG-I

162. Mr. G.Dasaratha, SSG-I
163. Ms. M.Eswaramma, SSG-I

Electrical Wing

164. Mr. L.Ramesh, Electrician cum Plumber (T-1-3)
165. Mr. K.Obulpathi, Technician Electrical (T-1)
166. Mr. M.K.Sonkusare, Electrician (T-1)
167. Mr. M.Mohan Rao, Technician Electrical (T-1)
168. Mr. B.S.N.Murthy, Technician Electrical (T-1)
169. Mr. M.K.Shamshuddin, Technician Electrical (T-1)
170. Mr. A.Nageswara Rao, Lineman (T-1)
171. Mr. B.Satyanarayana, Technician Electrical (T-1)
172. Mr. D.R.Sahadeva Rao, Pump Driver

APPOINTMENTS

- | | | |
|-----|---------------------------|--------------------------|
| 1. | Mr. B.Pandu Reddy | Finance & Accts Officer |
| 2. | Mr. Abhijit Bhattacharjee | Public Relations Officer |
| 3. | Mr. Sohail Ahmad Khan | Junior Engineer (Civil) |
| 4. | Mr. P.Mohan Singh | Computer Assistant |
| 5. | Ms. Savithri Srinivasan | Mess Supervisor |
| 6. | Mr. A.M.Venkateswarlu | Compounder |
| 7. | Ms. Radha Sujatha | Junior Stenographer |
| 8. | Ms. S.Shanthi | Junior Stenographer |
| 9. | Ms. K.K.Rukmani Ammal | Junior Stenographer |
| 10. | Ms. S.Sesha Sai | Junior Stenographer |
| 11. | Ms. V.Shailaja | Junior Stenographer |
| 12. | Mr. B.K.Venkatram | Pump Driver |
| 13. | Mr. M.Srinivasa Rao | Pump Driver |
| 14. | Mr. K.R.Ghanshyam | Junior Clerk |
| 15. | Mr. P.Venkatesh | Junior Clerk |
| 16. | Ms. Y.Gayathri | Junior Clerk |
| 17. | Mr. P.Bhaktavatsalam | Junior Clerk |
| 18. | Mr. T.Laxman | Driver |
| 19. | Ms. Nazia Parveen | SSG-I |

AGRIUNIS PROJECT (Supported by Department of Science and Technology)

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|-----|------------------|---------------------|
| 20. | Mr. M.Srinivas | Programmer |
| 21. | Mr. B.Chakrapani | Technical Assistant |
| 22. | Mr. C.Phani Raj | Technical Assistant |

PROMOTIONS

Sl.No.	Name	Designation	Promoted as
1.	Ms. L.Jhansi Laxmi	Sr. Steno	Sr.P.A
2.	Ms. G.Manibai	SSG-II	SSG-III
3.	Mr. B.Narayana	SSG-II	SSG-III
4.	Mr. B.H.Dharma Raj	SSG-II	SSG-III
5.	Mr. P.Swamy	SSG-I	SSG-II
6.	Mr. Phool Kumar	SSG-I	SSG-II
7.	Mr. Khalid	SSG-I	SSG-II
8.	Mr. M.Yadaiah	SSG-I	SSG-II
9.	Mr. P.Balraj	SSG-I	SSG-II
10.	Mr. G.V.Bikshapathi	SSG-I	SSG-I
11.	Mr. C.Bikshapathi	SSG-I	SSG-II
12.	Mr. M.Ashok	SSG-I	SSG-II
13.	Ms. B.Santhamma	SSG-I	SSG-II
14.	Mr. B.Subba Rao	SSG-I	SSG-II
15.	Ms. S.Shakuntala	SSG-I	SSG-II
16.	Ms. K.Kalavathi	SSG-I	SSG-II
17.	Mr. S.Jangaiah	SSG-I	SSG-II
18.	Mr. K.Satyanarayana	SSG-I	SSG-II

TRANSFERS

Sl.No.	Name & Designation	From	To
	Mr. M.Suresh Kumar Chief Administrative Officer	IVRI Izatnagar	NAARM Hyderabad
	Mr. K.K.Bajpai Chief Administrative Officer	NAARM Hyderabad	ICAR HQ New Delhi
	Mr. C.P.Mohan Kumar Finance and Accounts Officer	NAARM Hyderabad	SBI Coimbatore
	Mr. W.S.Bhatt Senior Clerk	NAARM Hyderabad	NRCS Hyderabad

DEPUTATIONS

S.No.	Name & Designation	From	To
	Dr. Devesh Kishore Principal Scientist	NAARM Hyderabad	IGNOU New Delhi
	Dr. K. Vidyasagar Rao Senior Scientist	NAARM Hyderabad	ICRISAT Hyderabad
	Ms. Girija Kumari Junior Stenographer	NAARM Hyderabad	DRR Hyderabad

RESIGNATIONS

1. Mr. M.Srinivas Programmer (AGRIUNIS)
2. Mr. P.P.Shastri Junior Stenographer

DEATH

1. Mr. Mohammed Nazeer Khan Driver



Faculty Centre and Laboratory-cum-Library Building of the Academy

